environmental constraints on the use of fossil fuels. Under the Paris Climate Change agreement, India has committed to reduce the carbon intensity of its economy by at least 45 per cent by 2030 compared to 2005 levels, in addition to ensuring that 50 per cent of its cumulative electric power installed capacity will be from non-fossil fuel-based energy resources. So far, wind, solar power and biofuels are the main renewable energy sources that India is banking on. But the game-changer could come in the form of Green Hydrogen, with India launching a mission to make the country a global hub for its production, use and export. Green Hydrogen is produced using the high school principle of electrolysis, which employs an electrical current to break down water molecules to hydrogen and oxygen using electrodes. Its costs are currently prohibitive, but big firms worldwide, including Indian giants like Reliance, Adani and L&T, are working towards lowering them.

And with the transport sector intrinsically linked to energy, India will have to master other critical technologies for it to make the leap to an advanced economy. Lithium batteries, for instance, are the best alternative to conventional lead acid batteries not only because they charge much faster but also because they can store higher amounts of energy in similar volume, thereby lasting longer before needing recharges. That is the reason why Li-ion have become the batteries of choice in Electric Vehicles (EVs), soon replacing petrol and diesel engines, which have serious carbon emission concerns. NITI Aayog estimates that there will be a 50-fold increase in the domestic demand for advanced chemical cell batteries by the end of this decade—both for EVs and also for storage on electricity grids using solar and wind power. Globally, China is the big boy of batteries, accounting for 70 per cent of the world's manufacturing capacity. India needs to move fast—it has discovered mineable lithium in Kashmir, but has yet to master the technology to manufacture such batteries at a commercial level. Especially if the goal is to make 30 per cent of all private cars and 80 per cent of two- and three-wheelers totally electric by 2030.

The other imperative for India is high-speed transport to transform mobility. In railways, for instance, Vande Bharat trains travel at a peak of 160 kmph. In comparison, Japan's Shinkansen or bullet train can travel at 320 kmph, or almost twice the speed. With India tying up with Japan for the 500-km Mumbai-Ahmedabad rail corridor, the first Indian Shinkansen to roll out in 2027 will cut travel time between these two commercial hubs from seven hours to a mere two. India will need many more such high-speed trains, but made indigenously to meet the massive demand for faster rail traffic.

When it comes to roads, Nitin Gadkari, the Union minister for transport, has done a commendable job of adding over 50,000 km length of highways in the nine years of the Modi government. However, the daily average of 350 km that Indian truckers traverse is still half of what their counterparts in the US are able to do. In addition to reducing tolls, India needs intelligently controlled highways that provide real-time data on snarl-ups and accidents and help unclog roads for speedy travel.

Aatmanirbharta, or self-reliance, is what India needs in its food and defence sectors too. There cannot be a greater irony than the fact that while the country is the largest pro-



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